

BIO516

Assignment 1

Marks: 10

Date: 10/02/2026

Time: 10 days

Answer all 4 questions

1. The electrical mobility of the sodium ion in water is $5.19 \times 10^{-4} \text{ cm}^2/\text{sec-V}$ at 25°C ; what is its diffusion coefficient in cm^2/sec ? **[1]**

2. Side I of a membrane is bathed by a 10-mM solution of AX and Side II is bathed by 100 mM AX; $T = 37^\circ\text{C}$, the solutions are ideal, and the ions are monovalent. Using either the Behn or constant-field solution, sketch the variation of diffusion potential as U_A/U_X varies from zero to infinity. **[3]**

3. The intracellular concentrations of the major ions in human erythrocytes are: 135 mM K, 17 mM Na, and 77 mM Cl; the concentrations in plasma are: 4 mM K, 138 mM Na, and 116 mM Cl. $T = 37^\circ\text{C}$. Chloride is in equilibrium across the cell membrane.

(a) What is the cell membrane potential?

(b) What is the Na/K permeability ratio?

[1.5 + 1.5 = 3]

4. Consider the system where the internal composition of the cell is 14 mM Na and 140 mM K, and the ambient solution is 142 mM Na and 4 mM K. $T = 37^\circ\text{C}$. Assume $U_{\text{Na}}/U_{\text{K}} = 0.05$. Now assume that increasing amounts of potassium replace the sodium in the external solution, while maintaining the total concentration of the two ions (Total concentration: 146 mM).

Plot the membrane potential as a function of external potassium concentration, from $C_{\text{K}}^{\text{II}} = 4 \text{ mM}$ ($C_{\text{Na}}^{\text{II}} = 142 \text{ mM}$) to $C_{\text{K}}^{\text{II}} = 140 \text{ mM}$ ($C_{\text{Na}}^{\text{II}} = 6 \text{ mM}$). **[3]**